Population Cycles in a Dimensionless Predator–Prey System

B. Fieldwork

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Abstract

We examine the classical predator–prey interaction after rescaling to dimensionless variables, where the prey density is x and the predator density is y. The oscillation period and the phase lag between the two populations follow from the trajectory starting at the field-observed initial state.

1 The dimensionless system

In dimensionless form the dynamics read

$$\dot{x} = x - xy, \tag{1}$$

$$\dot{y} = xy - y. \tag{2}$$

2 Field observations

The pond was seeded from the survey counts, giving the initial state $x(0) = 2$ and $y(0) = 0.5$. All trajectories reported below start from these values; the predator peak follows the prey peak by roughly a quarter cycle, as expected for this system.

3 Discussion

Small perturbations of the initial state preserve the closed orbits, which is the stabilising property exploited by the field census design.